PROBLEM VF17

Thin Plate in a Couette Channel

Original statement · VF17.pdf

Two channels in parallel | Forces and power | Energy balance

OFFICIAL PROBLEM STATEMENT

PROBLEM 3 (1 hour) A perfect liquid of constant density ρ and constant viscosity μ fills the space between two infinite parallel walls separated by a distance $2h$, that move parallel to themselves in the same direction with a constant velocity U. As shown in the figure, a very thin solid plate of length $L \gg h$ is placed at the center of the channel. In the regions far upstream and far downstream from the plate the liquid moves with uniform velocity U due to the motion of the confining walls, and the modified pressure takes uniform values $P(x \to -\infty) \to P_{-\infty}$ and $P(x \to \infty) \to P_{\infty}$, respectively.

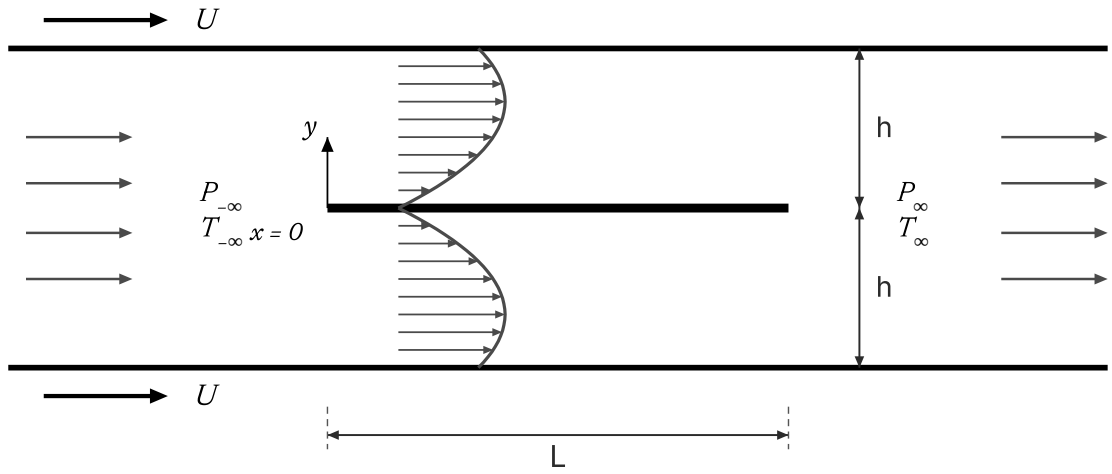
1. Provide the criterion that must be accomplished for the flow to be dominated by viscous forces. (**1 point**)

2. Obtain the velocity field in the region occupied by the plate, in terms of the unknown pressure gradient $P_l = -\partial P/\partial x$. (**3 points**)

3. Obtain the pressure drop $P_{-\infty} - P_{\infty}$, as well as the pressure gradient P_l. (**3 points**)

4. Determine the forces (per unit spanwise length) that the liquid exerts on the confining walls, F_w, and on the inner plate, F_p, as well as the power that must be applied to drive the walls. (**3 points**)

5. If all the inner walls are thermally isolated, determine the temperature increase in the liquid between two points situated far downstream and far upstream, $T_{\infty} - T_{-\infty}$. To that end, apply the energy equation in integral form to an appropriately defined control volume (**1 extra point**).



Statement figure (redrawn from VF17.pdf): a channel between two walls at y = ±h moving with velocity U, with a thin plate of length L on the centre line from x = 0 to x = L; the liquid far upstream and downstream moves with U at the pressures and temperatures P_{−∞}, T_{−∞} and P_∞, T_∞; the sketched profiles are those found in the solution

Source: VF17.pdf (page 1, exam of June 27, 2013; statement only, no key). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.8-7.12; slender flows, Eqs. 7.40-7.46.

Worked solution

Hypotheses, Data and Unknowns

- The walls are at $y = \pm h$ and move with U; the plate lies on $y = 0$ for $0 < x < L$ and is at rest. Over the plate the channel is divided into two identical half-channels of height h, symmetric with respect to $y = 0$.
- Steady flow, slender ($L \gg h$) and two-dimensional; the entrance and exit regions, of length of order h, are neglected: the pressure losses in them are small against the loss along the plate (Notes.pdf, Eq. 7.63).
- The liquid has constant c (specific heat), and the inner walls are insulated, so there is no heat flux across the boundaries. Per unit spanwise length.

Symbol	Meaning	SI unit
h, L	half-height of the channel, length of the plate	m
U	velocity of the walls and of the far flow	m/s
$P_{-\infty}, P_\infty$	reduced pressures far upstream and downstream	Pa

Symbol	Meaning	SI unit
$T_{-\infty}, T_{\infty}, c$	temperatures far away, specific heat	K, J/(kg K)

Criterion for Viscous Dominance

In the half-channel of height h and length L with the velocity scale U (Eq. 7.41):

$$\frac{\rho U h}{\mu}\,\frac{h}{L} \ll 1$$

The flow is steady. (If the pressure-driven velocity, computed below, is of the order of U the same criterion holds.)

Velocity Field in the Plate Region

In the upper half-channel, $0 < y < h$, with the plate at rest ($y = 0$) and the wall moving with U ($y = h$), the profile is Couette plus Poiseuille (Eq. 7.12):

$$v_x = U\,\frac{y}{h} + \frac{P_l}{2\mu}\,y\,(h - y), \qquad q_{half} = \int_0^h v_x\,dy = \frac{Uh}{2} + \frac{P_l h^3}{12\mu}$$

The lower half is the mirror image, $v_x(-y) = v_x(y)$.

Pressure Drop and Pressure Gradient

The flux is conserved along the channel. Far from the plate it is $2h\,U$ (uniform velocity). Over the plate it is the sum of the two halves:

$$2hU = 2\left[\frac{Uh}{2} + \frac{P_l h^3}{12\mu}\right] = Uh + \frac{P_l h^3}{6\mu} \implies P_l = \frac{6\mu U}{h^2}$$

The couette flux alone (Uh) is only half of what must pass, so the pressure must push: $P_l > 0$, the pressure falls in the direction of the flow. The gradient is uniform (the section is constant), so

$$P_{-\infty} - P_{\infty} = P_l L = \frac{6\mu U L}{h^2}$$

(the losses at the ends of the plate are of order $\mu U/h$, a factor h/L smaller). With this value the velocity is

$$v_x = U\frac{y}{h} + \frac{3U}{h^2}\, y\,(h-y) = \frac{U\,y}{h^2}\,(4h - 3y), \qquad 0 \le y \le h$$

which is U at the wall, zero at the plate and has a maximum $\frac{4}{3}U$ at $y = \frac{2}{3}h$ (the jet seen in the sketch of the sheet).

Forces and Power

The shear stress is $\mu\,\partial v_x/\partial y = \mu\left[\frac{U}{h} + \frac{P_l}{2\mu}(h-2y)\right] = \frac{\mu U}{h^2}\,(4h-6y)$. The force of the liquid on a surface is $\bar{\tau}' \cdot \bar{n}$ with $\bar{n}$ pointing from the surface into the liquid.

Walls. On the upper wall ($y = h, \bar{n} = -\bar{e}_y$): $\tau = -\mu \frac{\partial v_x}{\partial y}\Big|_h = -\frac{\mu U}{h^2}(4h - 6h) = \frac{2\mu U}{h}$ in $+x$: the liquid in the gap moves faster than the wall and drags it forward. Over the length L, for each wall (the walls outside the plate region are in uniform flow, with no shear):

$$F_w = \frac{2\mu U L}{h} \text{ on each wall,} \quad \text{in the direction of its motion;} \qquad \text{both walls: } \frac{4\mu U L}{h}$$

Plate. On each face ($y = 0^{\pm}, \bar{n} = \pm\bar{e}_y$): $\tau = \mu \frac{\partial v_x}{\partial y}\Big|_0 = \frac{4\mu U}{h}$ along $+x$. Two faces and the length L:

$$F_p = 2 \cdot \frac{4\mu U}{h}\, L = \frac{8\mu U L}{h}$$

(the drag on the plate, in the direction of the flow).

Check by momentum. For the liquid in the plate region (no change of momentum flux, uniform U in and out), the forces on it add to zero: the pressure difference on the section $2h$, $2h(P_{-\infty} - P_{\infty}) = 12\mu UL/h$, minus the reactions on the two walls, $4\mu UL/h$, minus the reaction on the plate, $8\mu UL/h$: $12 - 4 - 8 = 0$.

Power. The liquid pushes the walls forward with the total force $4\mu UL/h$, so to keep them at the constant velocity U an external force $-F_w$ must hold each one back. The power that the external agent supplies to the walls is

$$\dot{W}_{walls} = -2F_w U = -\frac{4\mu U^2 L}{h}$$

negative: the walls do not have to be driven, they **receive** the power $4\mu U^2 L/h$ from the liquid (which is being pushed through the plate region by the imposed pressure drop); a brake has to absorb it. Its magnitude is $2F_w U$.

Temperature Increase

Control volume: the liquid between two sections far upstream and far downstream, bounded by the walls and the plate (all insulated: no heat). Steady energy balance in integral form: net outflow of $\rho(e + v^2/2)$ plus the work of the pressure at the two sections equals the work of the viscous forces on the moving boundaries (the plate is at rest, the walls move with U). The kinetic energy flux is the same in and out (uniform U); with $e = cT$:

$$\rho c\,(2hU)\,(T_{\infty} - T_{-\infty}) = \underbrace{2hU\,(P_{-\infty} - P_{\infty})}_{\text{work of the pressure}} - \underbrace{2F_w U}_{\text{work done on the walls}} = \frac{12\mu U^2 L}{h} - \frac{4\mu U^2 L}{h} = \frac{8\mu U^2 L}{h}$$

(the work of the force that the walls exert on the liquid is $-F_w U$ per wall; the reduced pressure is used, since the flow is horizontal and the heights are the same). Hence

$$T_{\infty} - T_{-\infty} = \frac{8\mu U^2 L/h}{2\rho c\, hU} = \frac{4\mu UL}{\rho c\, h^2}$$

Check with the dissipation: $\Phi = \mu(\partial v_x/\partial y)^2 = \mu U^2(4h-6y)^2/h^4$, and $\int_0^h \Phi\, dy = \frac{\mu U^2}{h^4}\left[-\frac{(4h-6y)^3}{18}\right]_0^h = \frac{\mu U^2}{h^4}\,\frac{64h^3+8h^3}{18} = \frac{4\mu U^2}{h}$; two halves and the length L: $\frac{8\mu U^2 L}{h}$, the same, as it must be (all the mechanical power that is not stored becomes heat).

Answers and Checks

- Viscous dominance: $(\rho U h/\mu)(h/L) \ll 1$.
- Velocity over the plate: $v_x = \frac{U\,y}{h^2}(4h-3y)$ for $0 \le y \le h$ (symmetric in y); $P_l = 6\mu U/h^2$; $P_{-\infty} - P_{\infty} = 6\mu UL/h^2$.
- Forces: $F_w = 2\mu UL/h$ on each wall (forward), $F_p = 8\mu UL/h$ on the plate (forward). Power at the walls: $2F_w U = 4\mu U^2 L/h$, received from the liquid.
- Temperature: $T_{\infty} - T_{-\infty} = 4\mu UL/(\rho c h^2)$.
- Dimensions: $\mu UL/h$ per unit width is MT^{-2}, a force per unit width; $\mu UL/(\rho c h^2)$ is $ML^{-1}T^{-1}\cdot LT^{-1}\cdot L/(ML^{-3}\cdot L^2T^{-2}\Theta^{-1}\cdot L^2) = \Theta$.
- Limit $L \to 0$: all the quantities vanish, as they should without a plate.
- Check against the official key: none; the sheet has no solution, so all the results of this note are computed here and checked by the momentum, flux and energy balances above.